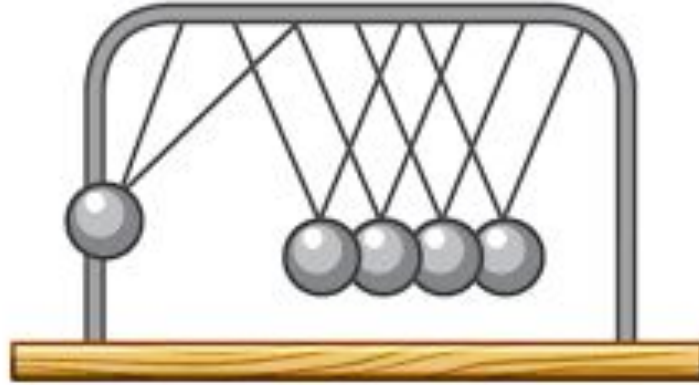




TU Delft Aerospace Engineering

Mechanics

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Kinematics

Kinematics describes motion through the relationship between position (x), velocity (v), and acceleration (a). These are linked by time (t).

The Relationships:

- Velocity is the rate of change of position: $v = dx/dt$
- Acceleration is the rate of change of velocity: $a = dv/dt$

Kinematics

A particle moves along the x-axis. Its position as a function of time t (in seconds) is given by the equation: $x(t) = -1/12 * t^4 + t^3 + 2t$

- 1) Maximum Acceleration: Find the time t at which the acceleration is at its maximum, and calculate the value of that maximum acceleration ($a_{\max}$).
- 2) Velocity at $a_{\max}$: What is the velocity of the particle at the exact moment it reaches its maximum acceleration?
- 3) Maximum Velocity: Determine the maximum velocity ($v_{\max}$) reached by the particle during the 10-second interval.
- 4) Overall Displacement: Calculate the total displacement of the particle after 10 seconds.

To solve these, we first need the first and second derivatives of $x(t)$:

- Velocity $v(t)$:

$$v(t) = \frac{dx}{dt} = \frac{d}{dt} \left(-\frac{1}{12}t^4 + t^3 + 2t \right) = -\frac{1}{3}t^3 + 3t^2 + 2$$

- Acceleration $a(t)$:

$$a(t) = \frac{dv}{dt} = \frac{d}{dt} \left(-\frac{1}{3}t^3 + 3t^2 + 2 \right) = -t^2 + 6t$$

Find Maximum Acceleration (a_{max})

To find the maximum of the acceleration function $a(t) = -t^2 + 6t$, we find where its derivative is zero:

- $a'(t) = -2t + 6$
- Set $a'(t) = 0 \Rightarrow -2t + 6 = 0 \Rightarrow t = 3$ s.
- Plug $t = 3$ back into $a(t)$: $a(3) = -(3)^2 + 6(3) = -9 + 18 = 9 \text{ m/s}^2$.

3. Velocity at the time of $a_{\max}$

We use the time found in the previous step ($t = 3$ s) and plug it into the velocity equation:

$$v(3) = -\frac{1}{3}(3)^3 + 3(3)^2 + 2$$
$$v(3) = -9 + 27 + 2 = 20 \text{ m/s.}$$

4. Find Maximum Velocity ($v_{\max}$)

Maximum velocity occurs when acceleration is zero ($a(t) = 0$):

$$-t^2 + 6t = 0 \Rightarrow -t(t - 6) = 0.$$

This happens at $t = 0$ (minimum) and $t = 6$ s (maximum).

Plug $t = 6$ into $v(t)$:

$$v(6) = -\frac{1}{3}(6)^3 + 3(6)^2 + 2$$
$$v(6) = -\frac{216}{3} + 3(36) + 2 = -72 + 108 + 2 = 38 \text{ m/s.}$$

5. Overall Displacement after 10 seconds

Since the particle starts at $x(0) = 0$, the displacement is simply $x(10)$:

$$x(10) = -\frac{1}{12}(10)^4 + (10)^3 + 2(10)$$
$$x(10) = -\frac{10,000}{12} + 1,000 + 20$$
$$x(10) = -833.33 + 1,020 = 186.67 \text{ m.}$$

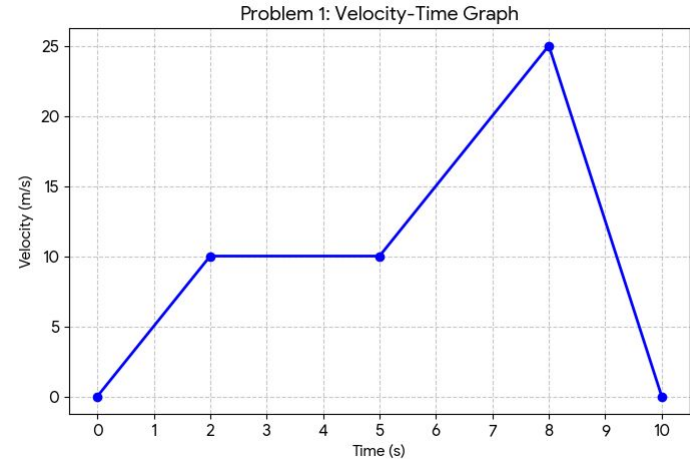
Diagram Analysis

We can visualize these variables in a diagram:

- 1) (x, t) graph: The slope represents velocity.
- 2) (v, t) graph: The slope represents acceleration; the area under the curve represents displacement.
- 3) (a, t) graph: The area under the curve represents the change in velocity.

Diagram Analysis

The graph represents the velocity of a particle moving along a straight line over a 10-second period.



- 1) Distance at $t = 5\text{s}$: Calculate the total distance traveled by the object during the first 5 seconds.
- 2) Total Distance: Calculate the total distance traveled by the object during the entire 10-second interval.
- 3) Maximum Acceleration: During which time interval is the magnitude of the acceleration the highest? Calculate the value of the acceleration in that interval.
- 4) Motion Description: Describe the state of motion (e.g., constant velocity, uniform acceleration) for the intervals $[2, 5]$ and $[8, 10]$.

Diagram Analysis:

Problem 1 Solutions:

1. **Distance at $t = 5\text{s}$:** The area under the $v - t$ curve from 0 to 5 seconds.

- Interval $[0, 2]$ (Triangle): $\frac{1}{2} \cdot 2\text{ s} \cdot 10\text{ m/s} = 10\text{ m}$.
- Interval $[2, 5]$ (Rectangle): $3\text{ s} \cdot 10\text{ m/s} = 30\text{ m}$.
- Total: $10 + 30 = 40\text{ m}$.

2. **Total Distance (0 – 10s):**

- Interval $[5, 8]$ (Trapezoid): $\frac{10+25}{2} \cdot 3\text{ s} = 17.5 \cdot 3 = 52.5\text{ m}$.
- Interval $[8, 10]$ (Triangle): $\frac{1}{2} \cdot 2\text{ s} \cdot 25\text{ m/s} = 25\text{ m}$.
- Grand Total: $40 + 52.5 + 25 = 117.5\text{ m}$.

3. **Maximum Acceleration:** Acceleration is the slope of the $v - t$ graph.

- Slope $[8, 10] = \frac{0-25}{10-8} = -12.5\text{ m/s}^2$.
- The highest magnitude is 12.5 m/s^2 (deceleration).

4. **Descriptions:**

- $[2, 5]\text{s}$: Constant velocity (10 m/s , $a = 0$).
- $[8, 10]\text{s}$: Constant deceleration (slowing down to a stop).

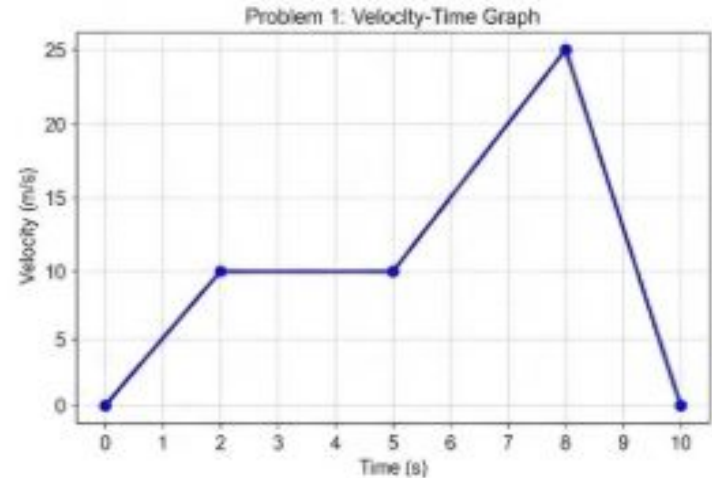
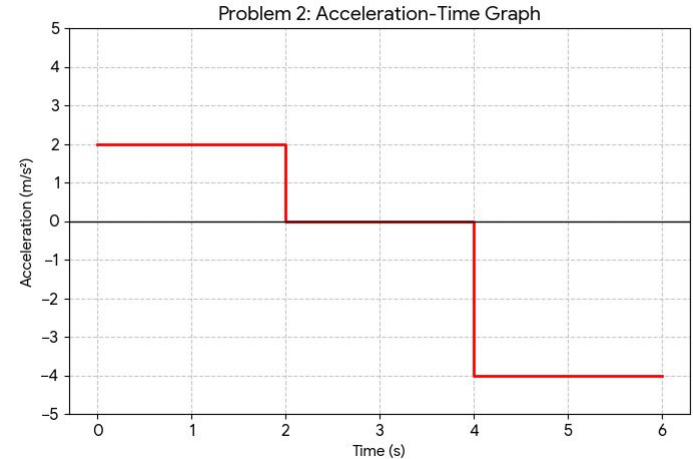


Diagram Analysis

An object starts from rest ($v_0 = 0 \text{ m/s}$) at the origin ($x_0 = 0 \text{ m}$). Its acceleration over time is shown in the graph.



- 1) Velocity-Time Graph: Sketch the corresponding (v , t) graph. Determine and label the velocity values at $t = 2\text{s}$, $t = 4\text{s}$, and $t = 6\text{s}$.
- 2) Position-Time Graph: Sketch the corresponding (x , t) graph. Identify where the graph is linear and where it is parabolic.
- 3) Final Displacement: What is the final position (displacement from the origin) of the object at $t = 6\text{s}$?
- 4) Turning Point: At what time t does the object reach its maximum positive displacement before potentially moving backward?

Diagram Analysis:

Problem 2 Solutions:

1. v-t values:

- $t = 2\text{ s}$: $v = \text{area of } a - t = 2 \text{ m/s}^2 \cdot 2 \text{ s} = 4 \text{ m/s}$.
- $t = 4\text{ s}$: $a = 0$, so velocity remains 4 m/s .
- $t = 6\text{ s}$: Change in $v = -4 \text{ m/s}^2 \cdot 2 \text{ s} = -8 \text{ m/s}$. Final $v = 4 - 8 = -4 \text{ m/s}$.

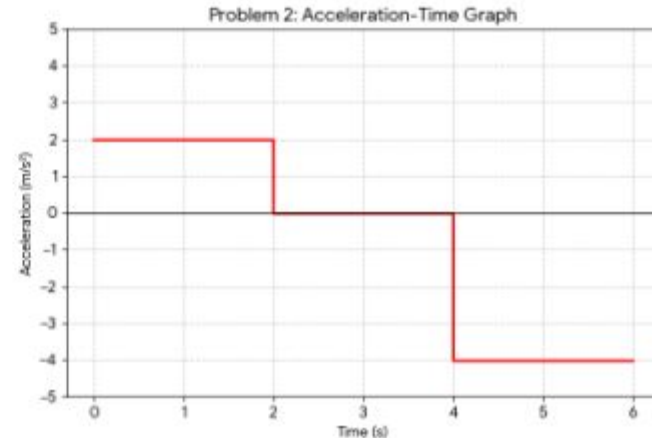
2. x-t graph:

- $[0, 2]\text{ s}$: Parabolic (concave up).
- $[2, 4]\text{ s}$: Linear (constant positive slope).
- $[4, 6]\text{ s}$: Parabolic (concave down).

3. Final Displacement: Area under the v - t graph you just derived.

- Area $[0, 2] = 4 \text{ m}$; Area $[2, 4] = 8 \text{ m}$; Area $[4, 5] = 2 \text{ m}$; Area $[5, 6] = -2 \text{ m}$.
- Total: $4 + 8 + 2 - 2 = 12 \text{ m}$.

4. Turning Point: The object moves forward as long as $v > 0$. The velocity crosses zero at $t = 5\text{ s}$, which is the point of maximum displacement (14 m).



Dynamics: Forces and Newton's Laws

Dynamics explains motion by looking at the forces (F) acting on an object.

Newton's Laws:

- 1) Inertia: An object stays at rest or in constant motion unless acted upon by a net force.
- 2) $F = ma$: Force equals mass times acceleration.
- 3) Action-Reaction: For every force, there is an equal and opposite reaction force.

Dynamics: Forces and Newton's Laws

Dynamics explains motion by looking at the forces (F) acting on an object.

Common Forces:

- Gravity: $F_g = m \cdot g$
- Springs (Hooke's Law): $F_s = k \cdot L$
- Friction: Opposes motion ($F_f = \mu \cdot F_N$)

Free-Body Diagrams (FBD):

A crucial tool where you draw all force vectors acting on a single point to "balance" them (set the sum of $F = 0$ for equilibrium or $\sum F = ma$ for motion).

Dynamics: Forces and Newton's Laws

An aircraft is currently in a steady climb. The following parameters are measured:

Mass (m): 10,000 kg

Thrust (T): 40,000 N (aligned with the aircraft's longitudinal body axis)

Drag (D): 8,000 N (acting opposite to the velocity vector)

Lift (L): 95,000 N (acting perpendicular to the velocity vector)

Flight Path Angle (γ): 15° (angle between the velocity vector and the horizon)

Pitch Angle (θ): 20° (angle between the aircraft nose and the horizon)

Dynamics: Forces and Newton's Laws

An aircraft is currently in a steady climb.

- 1) Draw the Free-Body Diagram (FBD): Show all forces (T , L , D , W).
- 2) Draw the Kinetic Diagram: Show the acceleration components (a_t and a_n).
- 3) Solve for Accelerations: Calculate the tangential acceleration (a_t) along the flight path and the normal acceleration (a_n) perpendicular to it.

Dynamics: Forces and Newton's Laws

Tangential Direction (a_t):

$$\sum F_t = T \cos(\alpha) - D - W \sin(\gamma) = ma_t$$

$$40,000 \cos(5^\circ) - 8,000 - 25,390 = 10,000 \cdot a_t$$

$$39,848 - 8,000 - 25,390 = 10,000 \cdot a_t$$

$$6,458 = 10,000 \cdot a_t \implies a_t = \mathbf{0.646 \text{ m/s}^2}$$

Normal Direction (a_n):

$$\sum F_n = L + T \sin(\alpha) - W \cos(\gamma) = ma_n$$

$$95,000 + 40,000 \sin(5^\circ) - 94,758 = 10,000 \cdot a_n$$

$$95,000 + 3,486 - 94,758 = 10,000 \cdot a_n$$

$$3,728 = 10,000 \cdot a_n \implies a_n = \mathbf{0.373 \text{ m/s}^2}$$

Work, Energy, and Power

This section deals with the currency of physics. Energy is conserved; it is never lost, only transformed.

Work (W): The product of force and displacement ($W = F * d * \cos(\theta)$).

Energy Types:

- Kinetic (E_k): $\frac{1}{2} * mv^2$ (Energy of motion)
- Potential (E_p): mgh (Energy of position)
- Elastic (E_{el}): $\frac{1}{2} * k * L^2$ (Energy stored in springs).

Work, Energy, and Power

This section deals with the currency of physics. Energy is conserved; it is never lost, only transformed.

Power (P): The rate of doing work ($P = W/t$)

Efficiency (η): The ratio of useful energy output to total energy input
($\eta = E_{\text{out}}/E_{\text{in}} * 100\%$)

Work, Energy, and Power

A jet engine is tested under the following flight conditions:

Airspeed (v): 250 m/s

Jet Exhaust Velocity (v_j): 600 m/s

Fuel Flow (m_f): 1.2 kg/s

Lower Heating Value (LHV) of Kerosene: 43 MJ/kg

Thrust (F): Assume for this problem the net thrust is 28,000 N.

- 1) Calculate Fuel Power (P_{fuel}): The total chemical power entering the engine.
- 2) Calculate Propulsive Power (P_p): The useful power used to move the aircraft.
- 3) Calculate Total Efficiency (η_{total}): The ratio of useful power to fuel power.

Work, Energy, and Power

Efficiency Calculation & Comparison

- **Fuel Power** (P_{fuel}): $P_{\text{fuel}} = \dot{m}_f \times LHV = 1.2 \text{ kg/s} \times 43,000,000 \text{ J/kg} = 51,600,000 \text{ W}$ (or 51.6 MW).
- **Propulsive Power** (P_p): $P_p = F \times v = 28,000 \text{ N} \times 250 \text{ m/s} = 7,000,000 \text{ W}$ (or 7.0 MW).
- **Total Efficiency** (η_{total}): $\eta_{\text{total}} = \frac{P_p}{P_{\text{fuel}}} = \frac{7,000,000}{51,600,000} \approx 0.1356$. $\eta_{\text{total}} = 13.56\%$.

Is this efficient? How does it compare to modern engines?

Engine Type	Typical η_{total}	Context
Early Turbojets (1950s)	10% – 18%	High jet velocity, massive heat loss.
Problem 5 Engine	13.56%	Performs like an early, low-bypass jet.
Modern Turbofans (e.g., GE9x)	30% – 40%	High bypass ratios and extreme temperatures.
Theoretical Limit	~60%	Limited by thermodynamics (Carnot) and drag.

Circular Motion and Orbits

When objects move in a circle, they are constantly accelerating toward the center, even if their speed is constant.

Key Variables:

- **Period (T):** Time for one rotation
- **Frequency (f):** Rotations per second ($f = 1/T$)
- **Angular Velocity (ω):** $2\pi f$ or $(2\pi)/T$.

Circular Motion and Orbits

When objects move in a circle, they are constantly accelerating toward the center, even if their speed is constant.

- 1) **Centripetal Force** (F_c): The center-seeking force required to keep an object in a circular path ($F_c = (m \cdot v^2)/r$).
- 2) **Planetary Motion**: In orbits, the Gravitational Force ($F_g = G \cdot (m_1 m_2)/r^2$) acts as the Centripetal Force.

By setting them equal, you can calculate orbital speeds and periods.

Circular Motion and Orbits

A telecommunications company wants to place a satellite of mass $m = 2,000 \text{ kg}$ into a Geostationary Orbit (GEO). In this orbit, the satellite remains fixed over a single point on Earth's equator because its orbital period matches Earth's rotation.

Given Data:

Mass of Earth (M_E): $5.97 \cdot 10^{24} \text{ kg}$

Radius of Earth (R_E): $6,371 \text{ km}$

Gravitational Constant (G): $6.67 \cdot 10^{-11} \text{ Nm}^2/\text{kg}^2$

Orbital Period (T): 24 hours

Circular Motion and Orbits

Angular Velocity: Determine the angular velocity (ω) of the satellite.

Force Balance: Set up the equation where Gravitational Force (F_g) acts as the Centripetal Force (F_c).

Orbital Radius: Calculate the distance (r) from the center of the Earth to the satellite.

Altitude: Calculate the height (h) above the Earth's surface where the satellite must fly.

Circular Motion and Orbits

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{86,400} \approx 7.27 \times 10^{-5} \text{ rad/s}$$

$$\sum F = F_c \implies F_g = F_c$$

$$G \frac{M_E \cdot m}{r^2} = m \cdot \omega^2 \cdot r$$

$$G \cdot M_E = \omega^2 \cdot r^3$$

$$r^3 = \frac{G \cdot M_E}{\omega^2}$$

$$r^3 = \frac{(6.67 \times 10^{-11}) \cdot (5.97 \times 10^{24})}{(7.27 \times 10^{-5})^2}$$

$$r^3 \approx 7.54 \times 10^{22}$$

$$r = \sqrt[3]{7.54 \times 10^{22}} \approx 42,240,000 \text{ m (or 42,240 km)}$$

$$h = r - R_E$$

$$h = 42,240 \text{ km} - 6,371 \text{ km} = \mathbf{35,869 \text{ km}}$$